

Next Feeding Session Predictor

Design Document v3 — Mathematical Formulae Preview · Kit + Koen · 2026-06-28

This document shows how the key mathematical definitions of the predictor look when typeset in a LaTeX-style formula layout. All formulae are exact; the prose around them is excerpted from the full design document.

§1 The Energy Model

The baby's daily nutritional need is derived from the paediatric guideline of 150 ml of prepared formula per kilogram of body weight per 24 hours. This gives a constant hourly drain rate:

$$D_{\text{target}} = w_{\text{kg}} \cdot r \text{ (ml / 24 h)}$$

$$\lambda = D_{\text{target}} / 24 \text{ (ml / h, constant)}$$

where w_{kg} is the baby's weight in kg and $r=150$ ml/kg/day (default). For a 6.9 kg baby:
 $D_{\text{target}}=1035$ ml, $\lambda=43.1$ ml/h.

Prepared milk is used as the direct proxy for caloric energy: 1 ml of prepared milk = 1 unit of energy. The standard interval — the time for one preferred bottle to be "spent" at rate λ — is:

$$SI = m_{\text{bottle}} / \lambda \text{ (hours)}$$

For a 90 ml water / 100 ml prepared milk bottle at 43.1 ml/h: $SI \approx 2\text{h}19\text{m}$.

§2 The 24-Hour Intake Function

Rather than tracking an all-time running balance, the app tracks the total milk received within a rolling 24-hour window. The credit contributed by a bottle of volume m_i given at time t_i , evaluated at time T , is:

$$\text{credit } T-t_i \text{ } m_i = m_i \text{ if } T-t_i \leq 24 \text{ h} \quad \max(0, m_i - \lambda \cdot (T-t_i-24) \text{ if } T-t_i > 24 \text{ h}$$

The total smoothed 24-hour intake at time T is the sum over all feeds logged before T:

$$\text{intake}(T) = \sum_{i=1}^n \text{credit}(T - t_i) m_i$$

The decay rate past 24 h is exactly λ — the same rate at which the baby consumes energy. This choice is not arbitrary: it means that a bottle's contribution to the 24-hour window diminishes at exactly the speed at which its nutritional value is consumed. In steady-state feeding, the net change per interval is zero, and intake oscillates between D_{target} and $D_{\text{target}} + m_{\text{bottle}}$.

§3 Predictor A — Adjusted Volume at Standard Time

Predictor A answers: "if I feed at the standard time SI after the last feed, how much should I give?" It solves for the volume that restores intake to the equilibrium peak $D_{\text{target}} + m_{\text{bottle}}$.

$$TA = t_{\text{last}} + SI$$

$$VA = D_{\text{target}} + m_{\text{bottle}} - \text{intake}(TA)$$

VA is converted to water ml via the inverse of the water→milk table and shown to the parent. At equilibrium: $\text{intake}(TA) = D_{\text{target}}$, so $VA = m_{\text{bottle}}$ — the standard bottle.

If $VA \leq 0$, the baby is still above the equilibrium ceiling at the standard time — Predictor A is greyed out and Predictor B takes over. If $VA > m_{\text{cap}}$, the value is capped at the stomach capacity limit (§5).

§4 Predictor B — Standard Volume at Adjusted Time

Predictor B answers: "when should I give my preferred standard bottle?" It finds the time TB at which the 24-hour intake has decayed back to exactly the daily target, so that adding one standard bottle restores intake to the equilibrium peak:

$$\text{intake}(TB) = D_{\text{target}}$$

Feed when the 24-hour intake returns to D_{target} . Adding m_{bottle} then restores intake to $D_{\text{target}} + m_{\text{bottle}}$.

TB is found by binary search on the interval [tlast,tlast+Tmax], where Tmax is the search horizon (default 48 h). Three cases:

Condition	Result	Meaning
intake(tlast)≤Dtarget	TB=tlast	Feed now — baby is at or below target
intake(Tmax)>Dtarget	TB=Tmax	Still in surplus at horizon — cap applies
otherwise	binary search	until intake(TB)–Dtarget <1 ml

At equilibrium TB=tlast+SI — the standard rhythm. With a deficit, TB<tlast+SI; with a surplus, TB>tlast+SI.

§5 Stomach Capacity Constraint

The total undigested milk in the stomach at time T — the milk not yet metabolised — is modelled as:

$$\text{Sload}(T) = \sum_{i=1}^n \max(0, m_i - \lambda \cdot (T - t_i))$$

The constraint is that the stomach load immediately after a feed must not exceed the capacity mcap (one standard bottle size up from preferred). This gives a minimum elapsed time before the next feed:

$$\Delta t_{min} = \max(0, m_{last} + m_{bottle} - m_{cap}) / \lambda$$

For a 6.9 kg baby with 90 ml water (100 ml milk) preferred bottle and stomach cap of 135 ml:
 $\Delta t_{min} = (100 + 100 - 135) / 43.1 \approx 1\text{h } 30\text{m}$

If Predictor B's TB falls before tlast+Δtmin, it is clamped upward with a note: "Stomach not ready — baby is underfed but needs time to digest."